

LAB 6	
AIM:	To implement the T-test model
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Steps.:	1. Import the required packages 2. Create the two variable x & y 3. Assign the values to the 2 variable 4. Find the standardization of it 5. Print the result 6. End the result
Program.:	<pre> import numpy as np import seaborn as sns import matplotlib.pyplot as plt from scipy import stats np.random.seed(42) a = np.random.normal(80, 8, 30) b = np.random.normal(75, 10, 30) t_stat, p_val = stats.ttest_ind(a, b) print(f'Analysis Results:') print(f'{'-'*20}') print(f'T-statistic: {t_stat:.3f}') print(f'P-value: {p_val:.3f} ({'Significant' if p_val < 0.05 else 'Insignificant'})') plt.figure(figsize=(8, 5)) sns.kdeplot(a, fill=True, color='royalblue', label='University A (Mean=80)') sns.kdeplot(b, fill=True, color='seagreen', label='University B (Mean=75)') </pre>

Program:

```
plt.title('Technical Assessment: Score Distribution Comparison', pad=15)
plt.xlabel('Assessment Score')
plt.ylabel('Density')
plt.legend()
sns.despine()
plt.tight_layout()
plt.show()
```

Output:

